

NEWTON'S INTERPOLATION FORMULA AND SUMS OF POWERS

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ABSTRACT. In this manuscript, we derive formulas for multifold sums of powers by utilizing Newton's interpolation formula. Furthermore, we provide formulas for multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers.

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1. INTRODUCTION

2. AUXILIARY LEMMAS

For multifold sums of powers, we use the notation provided by Donald Knuth in [?], because it is simple and beautiful.

Definition 2.1 (Multifold sums of powers). *For non-negative integers r, n, m*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

We utilize the following notation for rising factorials

Definition 2.2 (Rising factorials). *For integers n, k*

$$n^{(k)} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ \prod_{j=0}^{k-1} (n + j), & \text{if } k > 0. \end{cases}$$

We encourage the audience to read J. F. Steffensen's work on the definition of generalized factorials [?].

Lemma 2.3 (Forward Newton's formula). *Let $f: \mathbb{Z} \rightarrow \mathbb{C}$ be a function, and let $t \in \mathbb{Z}$. Then, for all $n \in \mathbb{Z}$,*

$$f(n) = \sum_{k=0}^{\infty} \frac{(n-t)^{(k)}}{k!} \Delta^k f(t) = \sum_{k=0}^{\infty} \binom{n-t}{k} \Delta^k f(t),$$

where the k -th forward difference $\Delta^k f(t)$ of f evaluated at t is defined by

$$\Delta^k f(t) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} f(t+j).$$

Proof. By [?, section 2, eq. (19)], and by identity $\frac{(n-t)^{(k)}}{k!} = \binom{n-t}{k}$. □

Lemma 2.4 (Forward Newton's formula for powers). *For non-negative integers n, m , and an arbitrary integer t*

$$n^m = \sum_{k=0}^m \binom{n-t}{k} \Delta^k t^m.$$

Proof. Special case of Lemma (2.3) for powers. □

Lemma 2.5 (Generalized hockey-stick identity). *For arbitrary integers a, b and j ,*

$$\sum_{k=a}^b \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^b \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right)$. By hockey-stick identity $\sum_{k=0}^n \binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^b \binom{k}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. □

Lemma 2.6 (Forward hockey-stick identity). *For integer $n \geq 1$, and arbitrary integers t, k, r*

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Proof. By setting $a = 1 - t + \frac{k}{2} + r$, and $b = n - t + \frac{k}{2} + r$ into Lemma (2.5) yields

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

This completes the proof. □

Lemma 2.7 (Multifold sums of zero powers). *For non-negative integers r, n*

$$\Sigma^r n^0 = \binom{r+n-1}{r}.$$

Proof.

(1) Let $r = 0$, then we have $\Sigma^0 n^0 = 1$, by definition (2.1).

(2) Let $r = 1$, then we have $\Sigma^1 n^0 = \sum_{k=1}^n 1 = \binom{n}{1}$.

(3) Let $r = 2$, then we have $\Sigma^2 n^0 = \sum_{k=1}^n \binom{k}{1} = \binom{n+1}{2}$.

(4) Let $r = 3$, then we have $\Sigma^3 n^0 = \sum_{k=1}^n \binom{k+1}{2} = \binom{n+2}{3}$.

(5) Similarly, for arbitrary integer r , by hockey-stick identity $\sum_{k=1}^n \binom{k}{r} = \binom{n+1}{r+1}$, yields

$$\Sigma^r n^0 = \sum_{k=1}^n \Sigma^{r-1} k^0 = \sum_{k=1}^n \binom{k+r-2}{r-1} = \binom{r+n-1}{r}.$$

This completes the proof. $\square$

Convention 2.8. *For all x*

$$x^0 = 1.$$

Donald Knuth give extensive discussion on the convention $x^0 = 1$ for all x , including zero, in Concrete Mathematics [?, p. 162], and in [?].

3. MAIN RESULTS

Hence, by Newton's formula for powers (2.4), we get formula for ordinary sums of powers.

$$\Sigma^1 n^m = \sum_{k=1}^n \sum_{j=0}^m \binom{k-t}{j} \Delta^j t^m = \sum_{j=0}^m \Delta^j t^m \sum_{k=1}^n \binom{k-t}{j},$$

where t is an arbitrary integer. By shifted hockey stick identity (??), the closed form of ordinary sums of powers yields.

Proposition 3.1 (Ordinary power sums). *For integers $n \geq 0$, and $m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \sum_{j=0}^m \Delta^j t^m \left[\binom{n-t+1}{j+1} - \binom{1-t}{j+1} \right].$$

4. STIRLING FORM

5. EULERIAN FORM

6. BACKWARD DIFFERENCE FORM

7. CENTRAL DIFFERENCE FORM

8. CONCLUSIONS

9. ACKNOWLEDGEMENTS

10. MATHEMATICA

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